INCH-POUND

MIL-DTL-24640/4D
22 August 2002
SUPERSEDING
MIL-C-24640/4C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 20, TYPE TTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, Size 20 AWG.
Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.
Third - Two component wires, one black and one white, shall be twisted together to form the required number of pairs. The pitch shall be 1 ± ¼ inch. Identification codes applied using Method 6.
Fourth - The required number of pairs shall be cabled together
Fifth - Binder tape applied helically with overlap, at manufacturer's option.
Sixth - Crosslinked polyolefin jacket. (See Table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Construction details.

PIN M24640/4	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
-01UN	TTX-3	3	0.038	0.296	0.320	62.1
-02UN	TTX-15	15	.044	.549	.591	221.0

REQUIREMENTS:

Qualification required.
Construction details: See Table I.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Color: In accordance with MIL-STD-104, class 1; white preferred
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 kHz spark test,
5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical: Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum.....10.95
Voltage withstand - volts, root mean square, minimum

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Conductor to conductor 2000
Conductor to shield ..... Not applicable
Shield to shield ..... Not applicable
Insulation resistance – megohms-1000 feet, minimum. 2500
Conductor continuity ..... No failure
Jacket flaws ..... Not applicable

Group A:
Visual and dimensional ..... No failure
Capacitance:
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum ..... Not applicable
Capacitance unbalance - percent, maximum ..... Not applicable
Characteristic impedance - at 1 MHz, ohms ..... Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage) ..... Not applicable
Weight ..... No failure

Group B:
Cold bending ..... No failure
Crosslink proof test (percent maximum)
Jacket 50
Drip ..... Not applicable
Tear - pounds per inch thickness, minimum 35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
Gas flame (1 hour) ..... Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - cycles, method 6 only, minimum 25
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage ..... Not applicable
Heat distortion - percent of unaged, maximum 30
Cable sealant removability Not applicable
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance
Milliohms per meter, maximum ..... Not applicable
EMP response, dB, minimum ..... Not applicable
Pulse response time - ms/500 feet
Rise time, maximum ..... Not applicable
Decay time, maximum ..... Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation	No failure
Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance No failure	
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component Wire	No Failure

Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-004)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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